

Exercises on Knowledge Representation and Reasoning in Propositional Logic

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Exercise 1

Which one among the following formulae is a good representation of the sentence?

If John studies and his father works, then his grandfather is happy.

- ① $Study \wedge Work \supset Happy$
- ② $Study \wedge Work \wedge Happy$
- ③ $\neg Study \vee \neg Work \vee Happy$
- ④ $Study \vee Work \supset Happy$

Exercise 1

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- ③ $\neg Study \vee \neg Work \vee Happy$
- ④ $Study \vee Work \supset Happy$

Solution:

- ① $Study \wedge Work \supset Happy$ - correct!
- ② $Study \wedge Work \wedge Happy$ - incorrect!
- ③ $\neg Study \vee \neg Work \vee Happy$ - correct! Why?
- ④ $Study \vee Work \supset Happy$ - incorrect

Exercise 2

Express the following knowledge

Considering the following propositional symbols:

- ST to indicate that *Pierino studies*;
- SY to indicate that *Pierino is silly*;
- LU to indicate that *Pierino is lucky*;
- PE to indicate that *Pierino passes the exam*.

Express the following knowledge:

- ① *If Pierino studies and is not silly, he passes the exam.*
- ② *If Pierino is not lucky and is silly, he does not pass the exam.*

Exercise 3

Express the following knowledge

Considering the following propositional symbols:

- ST to indicate that *Pierino studies*;
- SY to indicate that *Pierino is silly*;
- LU to indicate that *Pierino is lucky*;
- PE to indicate that *Pierino passes the exam*.

Express the following knowledge:

- 1 If *Pierino studies and is not silly, he passes the exam*.
- 2 If *Pierino is not lucky and is silly, he does not pass the exam*.

Solution: the knowledge is captured by the following knowledge base $\mathcal{KB}$:

- 1 $ST \wedge \neg SY \supset PE$
- 2 $\neg LU \wedge SY \supset \neg PE$

Exercise 4

Logical implication

Considering the following knowledge base $\mathcal{KB}$:

- ① $ST \wedge \neg SY \supset PE$
- ② $\neg LU \wedge SY \supset \neg PE$

Does the following logical implication hold?

$$\mathcal{KB} \models PE$$

Exercise 5

Abduction

Considering the following knowledge base $\mathcal{KB}$:

- ① $ST \wedge \neg SY \supset PE$
- ② $\neg LU \wedge SY \supset \neg PE$

Which set of formulas $\mathcal{AB}$ can be added to $\mathcal{KB}$ so that the following logical implication hold, while maintaining $\mathcal{KB} \cup \mathcal{AB}$ consistent? (Note this is a form of abduction.)

$$\mathcal{KB} \cup \mathcal{AB} \models PE$$

Exercise 6

Which one is a model?

Considering the following knowledge base $\mathcal{KB}$:

- ① $ST \wedge \neg SY \supset PE$
- ② $\neg LU \wedge SY \supset \neg PE$

which of the following interpretations is a model of $\mathcal{KB}$?

- $\{ST, SY, PE\}$
- $\{LU, PE\}$
- $\{\}$
- $\{LU, SY, PE\}$

Exercise 7

Knowledge base and logical implication

Considering the following knowledge:

- ① *If I leave and go on vacation, then I am happy*
- ② *If I leave then I go on vacation*
- ③ *I leave*

Express it in a knowledge base $\mathcal{KB}$. Then use the $\mathcal{KB}$ to answer the following query:

- *Will I go on vacation and be happy?*

Exercise 8

Knowledge base and inconsistency

Considering the following knowledge:

- ① *If it rains then it is wet*
- ② *If it is wet then it does not rain*
- ③ *It rains*

Express it in a knowledge base $\mathcal{KB}$. Then prove that the $\mathcal{KB}$ is *inconsistent*.

Exercise 9

Knowledge base and logical implication

Considering the following knowledge:

- ① *I'm happy iff I won the lottery or my girlfriend is with me*
- ② *If it is raining my girlfriend is not with me*
- ③ *It is raining and I am happy*

Express it in a knowledge base $\mathcal{KB}$. Then use the $\mathcal{KB}$ to answer the following query:

- *I am happy iff I won the lottery?*

Formalizing English sentences

Exercise

Let A = "Angelo comes to the party", B = "Bruno comes to the party", C = "Carlo comes to the party", and D = "Davide comes to the party".

Formalize the following sentences:

- ① "If Davide comes to the party, then Bruno and Carlo come too"
- ② "Carlo comes to the party only if both Angelo and Bruno do not come"
- ③ "If Davide comes to the party, then, if Carlo doesn't come then Angelo comes"
- ④ "Carlo comes to the party provided that Davide doesn't come, but, if Davide doesn't come, then Bruno doesn't come"
- ⑤ "A necessary condition for Angelo coming to the party, is that, if Bruno and Carlo aren't coming, Davide comes"
- ⑥ "Angelo, Bruno and Carlo all come to the party if and only if Davide doesn't come, but, if neither Angelo nor Bruno come, then Davide comes only if Carlo comes"

Formalizing English sentences

Exercise - Solution

- ① *"If Davide comes to the party then Bruno and Carlo come too"*
 $D \rightarrow B \wedge C$
- ② *"Carlo comes to the party only if Angelo and Bruno do not come"*
 $C \rightarrow \neg A \wedge \neg B$
- ③ *"If Davide comes to the party, then, if Carlo doesn't come then Angelo comes"*
 $D \rightarrow (\neg C \rightarrow A)$, which is equivalent to $(D \wedge \neg C) \rightarrow A$

Formalizing English sentences

Exercise - Solution

- ① *"Carlo comes to the party provided that Davide doesn't come, but, if Davide doesn't come, then Bruno doesn't come"*
 $(C \rightarrow \neg D) \wedge (\neg D \rightarrow \neg B)$
- ② *"A necessary condition for Angelo coming to the party, is that, if both Bruno and Carlo don't come, Davide comes"*
 $A \rightarrow (\neg B \wedge \neg C \rightarrow D)$
- ③ *"Angelo, Bruno and Carlo all come to the party if and only if Davide doesn't come, but, if neither Angelo nor Bruno come, then Davide comes only if Carlo comes"*
 $(A \wedge B \wedge C \leftrightarrow \neg D) \wedge (\neg A \wedge \neg B \rightarrow (D \rightarrow C))$

Formalizing English sentences

Exercise

Formalize the following arguments and verify whether they are correct:

- *"If you play and you study you'll pass the exams."*

If you play and don't study you won't pass.

Thus, if you play, either you study and you'll pass the exams, or you don't study and you won't pass."

Formalizing English sentences

Exercise

- *"If you play and you study you'll pass the exams."*

If you play and don't study you won't pass.

Thus, if you play, either you study and you'll pass the exams, or you don't study and you won't pass."

- - 1 $p \wedge s \rightarrow e$
 - 2 $p \wedge \neg s \rightarrow \neg e$
 - 3 $p \rightarrow (s \wedge e) \vee (\neg s \wedge \neg e)$

We need to prove that $(1) \wedge (2) \models (3)$